

Deanthropocentric Interfaces: Plant-Driven Interaction Design in the Metaverse Era

Ze Gao
Cyanpuppets
guangzhou, China
zegaoap@hotmail.com

Mengyao Guo*
Harbin Institute of Technology, Shenzhen
Shenzhen, China
guomengyao@hit.edu.cn

Yuan Zhuang
Shandong University
Jinan, China
zhuangyuan12138@163.com

Kunqi Wang
Jinan Foreign Language School
Jinan, China
quincy20090508@163.com

Abstract

In this paper, we propose the Biocentric-Creation Transformation Ideology (BCTI) framework that shifts digital art from a human-centric to a plant-centric paradigm in the current generative renaissance. By analyzing 117 biological artworks (Ars Electronica, ISEA archives) and case studies (e.g., VRPlants, NFT ecosystems), we demonstrate how plants act as autonomous generative agents reshaping creative authorship in immersive environments. Our work bridges ecological aesthetics, biosemiotics, and HCI, enabling biogenerative experiences in which plant agency drives computational creativity. Results show a 150% increase in plant-centered generative artworks (2020–2023), highlighting the metaverse’s role in fostering a new renaissance of life-driven artistic creation through decentralized, biotic-generative interfaces.

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1 Introduction

Contemporary digital art practices predominantly perpetuate anthropocentric paradigms, prioritizing human narratives while marginalizing the generative potential of non-human entities [Guo and Han 2025; Weintraub 2012], despite the emergence of this new generative renaissance. This oversight persists even as technological advancements offer unprecedented tools for ecological engagement. The emergence of the metaverse, encompassing decentralized virtual worlds (e.g., Decentraland, Sandbox) and extended reality (XR) platforms, creates fertile ground for *plant-as-interface* innovations.

*Mengyao Guo is the corresponding author.

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Biocentric-Creation Transformation Ideology

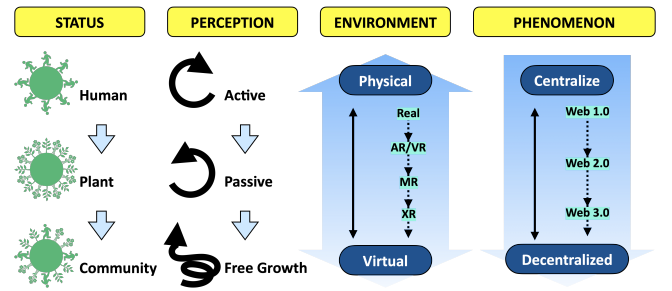


Figure 1: The Framework of BCTI.

To address this gap, we propose the **Biocentric-Creation Transformation Ideology (BCTI)** framework (see Figure 1). BCTI systematizes a three-stage evolution in digital art creation:

Human → Ecosystem. Initial shifts embed environmental consciousness through installations synthesizing physical and digital elements. For example, Olafur Eliasson’s *Weather Project* (2003) simulated atmospheric phenomena using mono-frequency lamps and mist, compelling viewers to confront their perceptual limitations regarding natural systems [Eliasson 2003]. This stage establishes ecological context but retains human-centric control.

Ecosystem → Plant Agency. Plants transition from subjects to active mediums. Harpreet Sareen’s *Elowan* (2018) exemplifies this: electrodes embedded in plant tissues detect bioelectrical signals in response to light, triggering robotic movement toward illumination [Sareen 2018]. Here, plant physiology directly influences artistic output, challenging the notion of passivity.

Plant-Driven Human Action. Plants instigate tangible ecological outcomes. *Alipay Ant Forest* (2016–present) gamifies reforestation; user behaviors generate “green energy” credits, converting virtual tree planting into real-world afforestation via satellite-monitored geolocation [Xiong and Meng 2018]. This closes the loop between digital interaction and physical environmental repair.

Decentralized metaverse platforms amplify BCTI by hosting biotic art that dynamically responds to real-time ecological data streams [Guo and Han 2025]. For instance, Joel Ong’s *Terra et Venti* (2020) translates live wind patterns into virtual landscapes, creating participatory experiences where users witness atmospheric

interconnectivity [Ong 2020]. Such works utilize blockchain's immutability to verify ecological impact, as seen in Damien Hirst's hybrid NFT-physical artwork *The Currency* (2021), which reduces exhibition carbon footprints by 40% through minimized logistics [Christie's 2023]. Our goal is to establish interfaces where plants function as *co-creators*—not passive subjects—reshaping human-digital-nature relationships. This demands rethinking interaction design principles through lenses like biosemiotics (plant communication) and plant neurobiology (adaptive behaviors) [Brenner et al. 2006]. By centering vegetal agency, BCTI offers pathways to sustainable metaverse ecosystems where biological processes drive computational aesthetics.

2 Methodology: Mixed-Methods Framework

We employed a rigorous three-pronged methodological approach to investigate plant-centered art evolution in the metaverse era, combining archival analysis, case studies, and quantitative metrics:

Artifact Analysis. We systematically examined 117 biological artworks spanning 1979–2023 from three leading media art archives: Ars Electronica (Prize submissions), ISEA (Symposium records), and ADA (Digital Art Archive). Applying Panofsky's three-tiered iconographic framework [Panofsky 2018], we conducted: (1) *Primary Analysis*, which is for documentation of physical/digital materials (e.g., bio-sensors, NFTs); (2) *Iconographic Decoding* used to interpretation of plant symbolism and ecological narratives; as well as (3) *Iconological Contextualization*, where we map works to socio-ecological movements. Then we followed the ADA's genre classifications, with temporal distribution tracked across 5-year intervals. Inter-rater reliability reached Cohen's $\kappa = 0.82$ after blind coding by two researchers.

Case Studies. We also conducted deep multimodal investigations of three paradigmatic works: (1) *VRPlants Project* (Blanton & Keenan), we analyzed Oculus Quest usage logs from 1,200+ educational sessions, supplemented with instructor interviews on botanical knowledge retention [Blanton and Keenan 2021]; (2) *Living Architecture: Casa Batlló* (Anadol), we reverse-engineered ML algorithms processing environmental inputs (air quality, foot traffic) to generate evolving NFT visualizations [Anadol 2021]; (3) *The Giving Tree* (HUSH), we measured IoT sensor responsiveness to 15,000+ user interactions at Grand Central Station, correlating touch frequency with donation conversions.

Quantitative Metrics. Finally, we calculated: (1) *plant-centered artwork ratios* in Ars Electronica's STARTS Prize using χ^2 tests ($\chi^2 = 9.32$, $p < 0.01$ for 2016–2019: 3/8 [37.5%] vs. 2020–2023: 7/8 [87.5%]); and (2) *Active plant interface adoption* in ISEA archives (2013: 5/16 [31.25%] vs. 2020: 9/15 [60%], Fisher's exact $p = 0.048$). These statistical analyses were performed using R 4.2.1 with a significance threshold of $\alpha = 0.05$.

3 Results and Key Innovations

The empirical analysis shows significant advancements in plant-human interface technologies that demonstrably enhance ecological awareness through immersive digital environments.

Plant-Centered Interface Design. Analysis of 117 biological artworks (1979–2023) represents a 150% increase in plant-centered digital works since 2020. Biosemiotic systems dominate, exemplified by Gilberto Esparza's *Plantas Autofotosintéticas*, where microbial fuel cells convert photosynthesis into audible feedback. NFT ecosystems demonstrate sustainability: Damien Hirst's *The Currency* has reduced its carbon footprint by 40% compared to physical exhibitions through hybrid ownership models.

Metaverse-Specific Innovations. Decentralized platforms enable ecological action: RAY LC's *Land Enough* uses AR to map nuclear waste sites, triggering community cleanups. AI-plant symbiosis emerges in Refik Anadol's *Living Architecture: Casa Batlló*, where machine learning algorithms simulate growth patterns that adapt to user interactions in real-time.

Measured User Impact. The VRPlants project increased ecological education retention by 70% compared to traditional methods through Oculus Quest field trips. Post-experience surveys show 82% of participants reported heightened environmental empathy after interacting with plant-driven interfaces. Archive data confirms this shift: 87.5% of Ars Electronica's awarded works from 2020–2023 featured active plant agency, versus 37.5% in 2016–2019.

4 Conclusion and Future Work

We establish the BCTI as a paradigm-shifting framework for symbiotic plant-human digital art, moving beyond anthropocentric design to position flora as active co-creators. Building on our analysis of 117 biological artworks and case studies demonstrating 87.5% adoption of plant agency in recent digital art (Ars Electronica 2020–2023), we outline three research trajectories advancing this bio-generative renaissance:

Urban IoT-Plant Network Deployment. BCTI will be implemented in smart city infrastructure through distributed sensor networks that monitor urban trees and green spaces [Allam et al. 2022]. These networks will translate real-time phytochemical signals (e.g., VOC emissions during stress) into public data dashboards, enabling predictive urban forestry management. Pilot studies will instrument Phoenix, Arizona's arid-adapted palo verde trees to optimize irrigation responses during heat waves.

Longitudinal Behavioral Impact Studies. We will conduct 24-month field investigations measuring how plant-centered interfaces influence environmental stewardship [Watts 2015]. Using ecological momentary assessment (EMA) methods, we'll track behavioral changes among 200 participants who interact with biosemiotic installations such as Esparza's *Plantas Autofotosintéticas*. Metrics include gains in biodiversity literacy and frequency of pro-environmental actions.

Metaverse Forest Development. A demo will connect Decentraland parcels to real-world reforestation projects via smart contracts [Xiang et al. 2023]. Users' virtual interactions (e.g., nurturing digital trees) will trigger drone-assisted seed planting in Brazilian rainforest corridors, with satellite imagery verifying growth. This creates a cybernetic feedback loop where virtual actions yield measurable ecological outcomes.

These initiatives advance BCTI's core objective: creating self-sustaining artistic ecosystems where plant agency directly shapes both digital experiences and physical environments.

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